

Model output report: full_PCA_LOF_k240

Summary of model_output_full_PCA_LOF_k240.csv (fusion of rule-based + full_PCA_LOF_k240 scores).

1. Overview

Item	Value
Total customers	61,410
Flagged (top 5%)	3,071 (5.00%)
Not flagged	58,339
Effective threshold (risk_score $\geq$)	0.3653

Insight: This model flags the top 5% of customers by combined risk score (rule-based plus full_PCA_LOF_k240 anomaly score), so 3,071 customers are marked for review. The threshold 0.3653 is the minimum score needed to be in that top 5%. All flagged customers should be prioritised for further investigation; the rest of the population is below the cutoff.

2. Risk score distribution

Statistic	All	Flagged	Not flagged
Min	0.0001	0.3653	0.0001
Max	0.8094	0.8094	0.3653
Mean	0.2035	0.4330	0.1914

Insight: The spread of scores (min to max, and standard deviation) shows how much the model separates customers. Flagged customers have higher mean and max scores than the rest; the gap between flagged and not-flagged means indicates how distinct the high-risk group is. IQR (interquartile range) is 0.144, so the middle 50% of scores span this range; a wider IQR suggests more variation in risk across the population.

3. Distribution by score band

Score band	Count	% of total
[0.0, 0.1)	12,217	19.9%
[0.1, 0.2)	13,631	22.2%
[0.2, 0.3)	26,716	43.5%
[0.3, 0.4)	7,176	11.7%
[0.4, 0.5)	1,176	1.9%
[0.5, 0.6)	302	0.5%
[0.6, 0.7)	167	0.3%

[0.7, 0.8)	23	0.0%
[0.8, 0.9)	2	0.0%
[0.9, 1.0)	0	0.0%

Insight: Most customers fall in the lower score bands; only a small share have scores in the 0.5+ range (0.8% here). The top 5% flagged set sits at the right tail of this distribution. If there are customers just below the threshold (within 0.02), there are 1,217 in that borderline band—these may warrant monitoring as small changes in behaviour could push them into the review set.

4. How to use this output

Use the corresponding `model_output_*.csv` and `model_output_*_explanations.csv` for Task 2 and Task 3. The explanations file gives a plain-English reason for each customer's score; the report above summarises the overall distribution and helps you understand how this model's fusion (rule + anomaly) behaves across the population.